

Learning for Data Science

Course 4: Complexity, Training Curve Diagnosis, Interpretability and Performance Metrics

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Université 
de Montréal



Mila

CIFAR

Outline

Complexity of a model in machine learning (ML)

Complexity of linear models

Learning and inference time

Interpretability

Diagnosis of training curves

Typical training plots of ‘healthy’ models

Classification

Classification metrics based on the area under a curve

One last Example

Complexity of a model in machine learning (ML)

- Ability to understand more and more information from the data, until it is completely “saturated” with information.
- Main factors affecting the complexity of a model
 - Number of parameters to train
 - structure/architectural
- A small deep neural network (NN) often performs better than a shallow but very large one.



Common situations when training a model in ML

- One can improve one's performance by increasing the training data.

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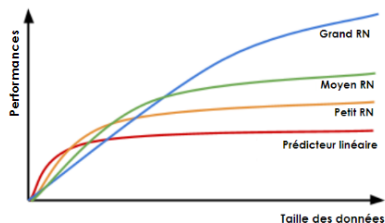
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- Maximum performance depends on model complexity.
- For better performance, increase model complexity or change the model.



Example of multilinear regression

- You can make a basic model more complex by adding new combinations of features:
 - Basic model:
 - Added interactions:
 - Added nonlinearities:
 - Added x attributes:
- The θ parameters of linear models are easy to estimate with the least squares method.
- The number of possible interactions increases as D^2 where D is the dimensionality of x .

$$\hat{y} = \theta_0 + \theta_1 x^1 + \theta_2 x^2$$

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$$\hat{y} = \theta_0 + \theta_1 x^1 + \theta_2 x^2 + \theta_3 x^1 x^2 + \theta_4 (x^1)^2 + \theta_5 (x^2)^2$$

$$\hat{y} = \theta_0 + \theta_1 x^1 + \theta_2 x^2 + \theta_3 x^3 + \dots$$

Complexity and decision trees

Models based on decision trees

- A decision tree implicitly manages the interactions between the attributes x without having to specify them as in a multilinear model.
- Making decisions trees more complex
 - random forest
 - Gradient Boosted Trees
 - Boosted Forests
 - XGBoost
- Many real world problems naturally lend themselves to analysis based on decision trees.

Example: modeling the profits of a ski resort

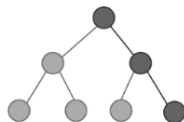
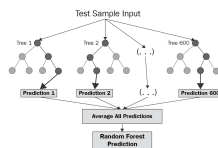
- We must take into account several attributes x
 - Weather report
 - Day of the week,
 - Leave?
 - heavy snow
- Each skier uses a decision tree with these attributes to decide whether or not to come skiing.
- Profits will therefore depend on the decisions of a large number of decision trees.



- The crowd decides like a random forest!

Strategies for controlling the complexity of a decision tree

- Reduced complexity
 - Limit its depth
- Increase in complexity
 - Switch to random forest



Adding layers to a neural network isn't always the answer

- It improves the performance of a neural network up to a certain point.

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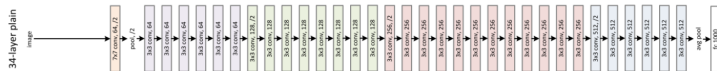
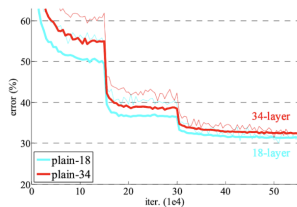
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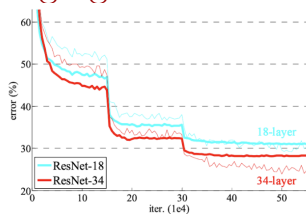
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- The classification error ends up increasing with the number of layers...

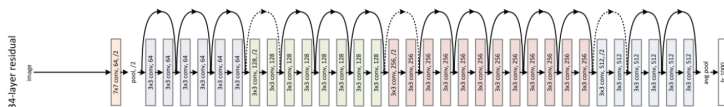


- Classic architecture of a convolutional RN

Changing Network structure is often more efficient



- The ResNet network surpasses the performance of conventional convolutional RNs.
- The structure of ResNet allows it to improve its performance by adding specific layers.
- Resnet model family (18, 34, 50, 101, 152 layers)



- Architecture of a ResNet network

Changing model families is even more efficient

- The performances eventually level off for a given structure family.

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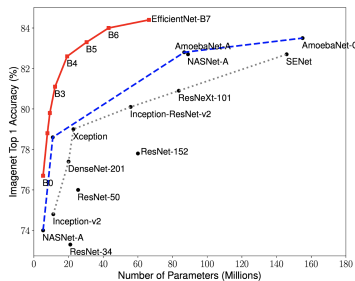
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- The performances eventually level off for a given structure family.
- It is necessary to test different structures of RN
- Too much tweaking of a preferred model decreases returns on performance.
- It is important to be aware of available literature and resources.



The importance of free software (Open Source code)

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- Several web hosting and software development management services exist.

statech/**resnet**

keras-style API to ResNets (ResNet-50, ResNet-101, and ResNet-152)



1
Contributor

58
Used by

22
Stars

20
Forks

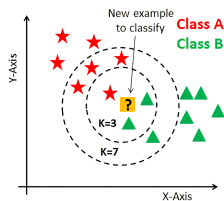


Inference time/latency: time required to make a prediction.

- Simple classification model: K nearest neighbors
 - Zero learning time!
 - Calculating distances to N training data each time...
 - Inference time proportional to K

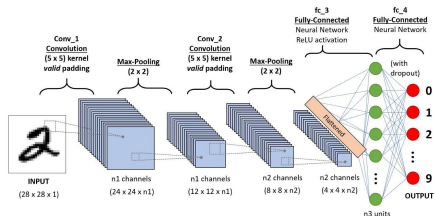
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 - Calculating distances to N training data each time...
 - Inference time proportional to K
 - Time in seconds if K is large
 - Inconvenient in real time...



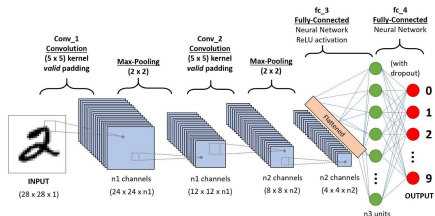
Complex classification model: neural network

- Long learning time (hours, days)...
- Can be reduced through learning transfer
- Constant inference time (a few ms to a second)
- Commonly used in real time!



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Machine Learning Time Complexity

DataInterview.com

n: data size
p: Number of features
T: Number of trees
l: Number of iterations
m: Number of components
h: Number of hidden units
k: number of clusters

ML Algorithms	Training Time	Inference Time
Linear Regression	$O(np^2 + p^3)$	$O(p)$
Logistic Regression	$O(np^2 + p^3)$	$O(p)$
Naive Bayes	$O(np)$	$O(p)$
Decision Tree	Avg: $O(T \cdot n \log n)$ Worst: $O(n^2)$	Avg: $O(T \cdot n \log n)$ Worst: $O(n)$
Random Forest	$O(T \cdot n \log n)$	$O(T \cdot \log n)$
Gradient Boosted Trees	$O(T \cdot n \log n)$	$O(T \cdot \log n)$
Principal Component	$O(np^2 + p^3)$	$O(pm)$
K-Nearest Neighbor	$O(1)$	$O(np)$
K-Means	$O(l \cdot k \cdot n \cdot p)$	$O(k \cdot p)$
Dense Neural Networks	$O(l \cdot n \cdot p \cdot h)$	$O(p \cdot h)$

Conclusion

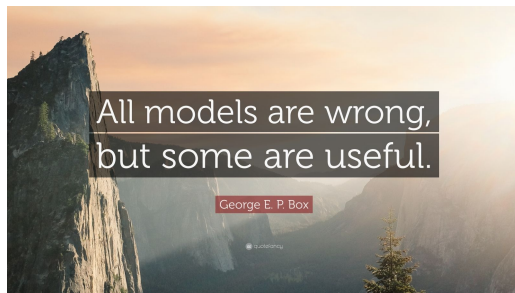
- Two main ‘use’ moments for models:
- Training:
 - Important when developing models
- Inference:
 - Important when using the model

Conclusion

- Two main ‘use’ moments for models:
- Training:
 - Important when developing models
- Inference:
 - Important when using the model
- Both can be reduced (requires more coding and resources) using:
 - Parallelization (distributed training)
 - A cluster.
 - More efficient hardware (GPU, vs CPU)
 - More details in dedicated courses.

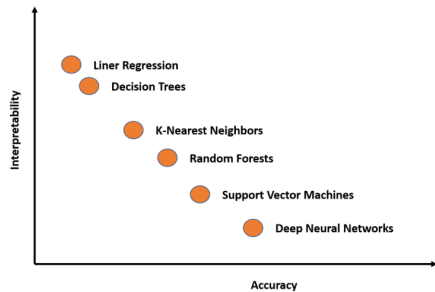
Choose a machine learning model

- AI provides many models for analyzing data.
- Some are precise, but incomprehensible.
- Others are simple, but imprecise (high bias).
- Each represents an approximation of reality.
- Which one to choose?



The importance of explaining the results of predictions

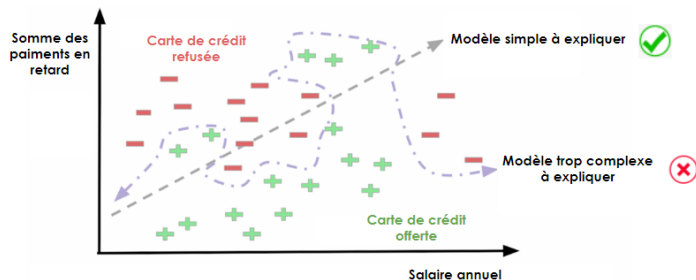
- In general, as the complexity of a model increases, its interpretability decreases.



- Complexity
 - Bank loan refused
 - Cancer detection
 - application declined
 - Etc.
- Concrete examples:

Sample credit card application

- How could a banker explain his decision to a client?
- Simple to explain



- Too complex to explain
- What is the 95% accuracy of the complex model, without confidence interval, and cannot be explained?

Interpretability can help detect bias

- Example of selection bias

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- Let's replace gender with ethnicity, and it will become a new primary attribute for the same reasons...

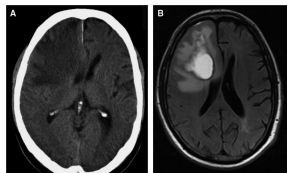
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- More details in the course on algorithmic/data biases.



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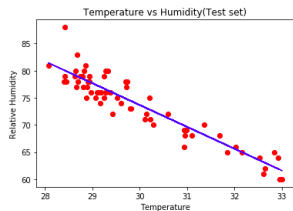
- Example of measurement bias
- A medical image analysis system is trained to detect cancers in MRI-type images.
- Problem:
- Positive cases are confirmed with machine B.
- The negative cases are all from machine A.
- The cancer detection rate will be biased by the machine used to get the image.



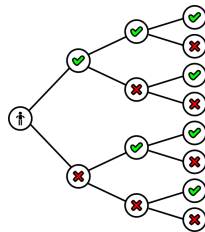
- CT scanner
- MRI

Interpretability of results: simple models

- It is easy to explain decisions made by simple models such as:
- Linear regression



- Decision Tree



Example of linear regression

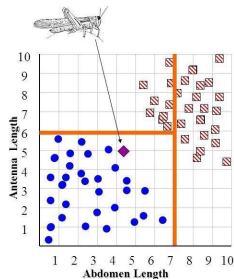
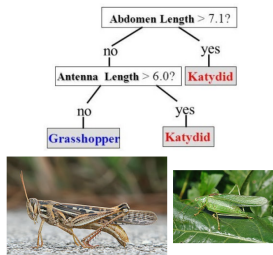
- General linear model:
 - Larger values $|\theta_i|$ correspond to more important (normalized) attributes x^i .
 - When $\theta_i > 0$, there is a positive correlation between the attribute and the response y .
 - A variation Δx^i produces a variation of the response $\Delta y = \theta_i \Delta x^i$ when the other attributes x^j remain unchanged.

Simple linear model: $y = \theta_0 + \theta_1 x$

Adding more attributes gives $y = \theta_0 + \theta_1 x^1 + \dots + \theta_D x^D$.

Example of a decision tree

- Decision tree fitted to the morphometric data (in cm) of two types of grasshoppers (grasshoppers and katydids in English)
- Only a few X attributes make it possible to differentiate the two types of insects.



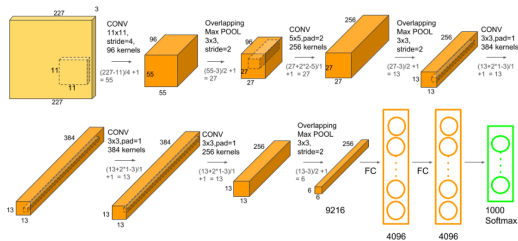
- Classification of a new grasshopper grasshopper.

Interpretability of results: neural networks

- To make predictions with a neural network, input data is passed through many layers performing matrix products followed by nonlinear transformations.
- Difficult to follow the flow of data from one layer to another.
- A single prediction can involve millions of mathematical operations depending on the architecture of the neural network.



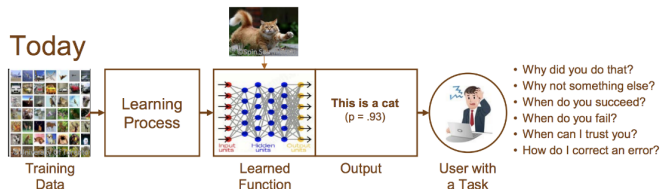
Example of a deep neural network of modest complexity



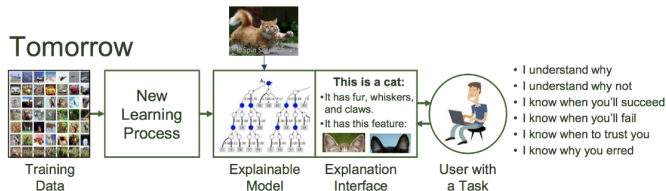
- The AlexNet convolutional neural network contains
- Eight hidden layers
- A nonlinear activation function (ReLU, Leaky ReLU, etc.)
- 62,378,344 parameters!

neural network interpretability is an active field of research

Today

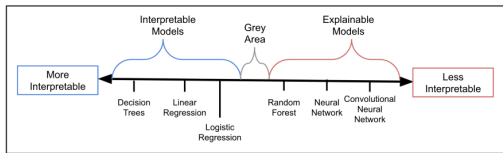


Tomorrow



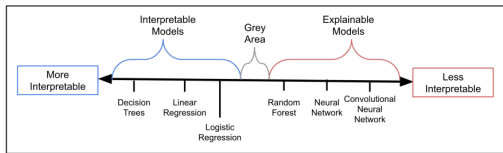
- It takes an expert to understand a result and its implications.
- GOAL: A new generation expert system will explain its decisions.

Differences between interpretable and explainable models



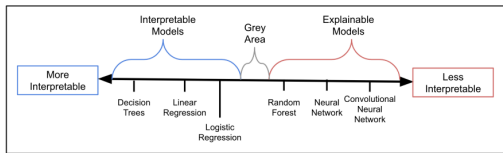
- An interpretable model is easy for a human to understand
- An explainable model cannot be understood exactly in a simple way, but there are strategies to analyze the factors that influence its predictions.
- Its structure is complex and the number of parameters is high.

Differences between interpretable and explainable models



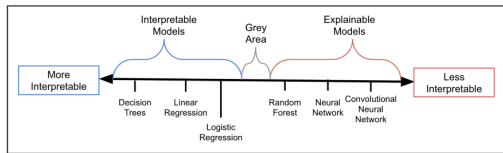
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What are the important factors in a model?

- Interpretable models often identify important factors
 - Linear regression
 - Logistic regression
 - Decision trees
- There are methods to explain locally important factors, i.e. for a particular datum
 - LIME (Ribeiro *et al.*, 2016)
 - SHAP (Lundberg and Lee, 2017)
 - DeepLIFT (Shrikumar *et al.*, 2017)

Example application of LIME for images

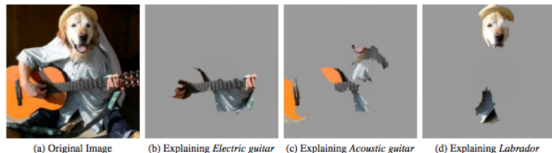


Figure 4: Explaining an image classification prediction made by Google's Inception network, highlighting positive pixels. The top 3 classes predicted are "Electric Guitar" ($p = 0.32$), "Acoustic guitar" ($p = 0.24$) and "Labrador" ($p = 0.21$)

- For the image above, LIME does this

Example application of LIME for images

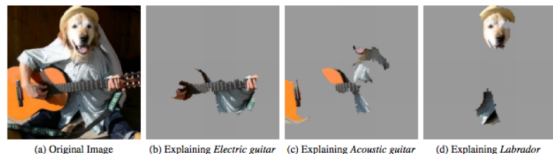


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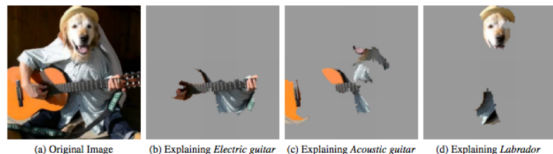


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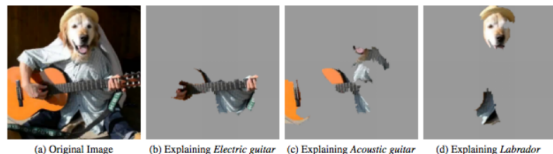
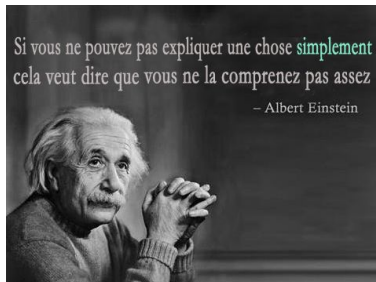


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- For the image above, LIME does this
- Determines the three most likely classes associated with the image.
- Identifies, in a given layer, the important x attributes (i.e. neurons)
- Hides areas in the image where these attributes have a negligible effect.

The importance of explaining a model

- If you can't explain one thing simply it means you don't understand it enough.



- This is true in relativity, in mathematics, in genomics, in biochemistry, and also in machine learning.

Diagnosis of training curves



- The ML stethoscope: the loss function curve

What is best for a newly trained classifier?

- These results based on two figures are difficult to interpret!
- Case A
 - 99% training accuracy
 - 80% validation accuracy
- Case B
 - 85% training accuracy
 - 80% validation accuracy
- Case C
 - 30% training accuracy
 - 25% validation accuracy



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- Overfitting



- optimal

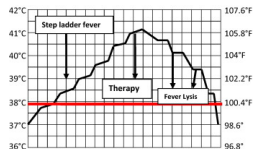


- Underlearning



The ML stethoscope

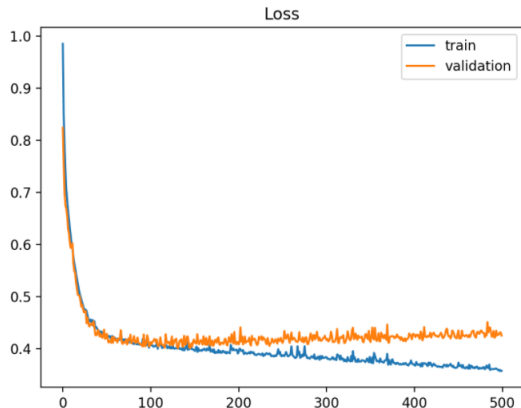
- For a doctor, the evolution of a patient's temperature is often more useful than the current temperature.



- It is not enough to be satisfied with the final performance of a model to decide if it is well trained; you should also see its different 'temperature plots':
 - The evolution of the loss function
 - The evolution of the performance metric

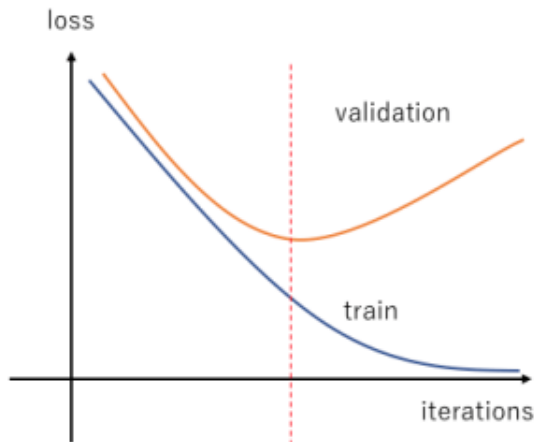
Learning plot:

- Plot of learning performance against time.

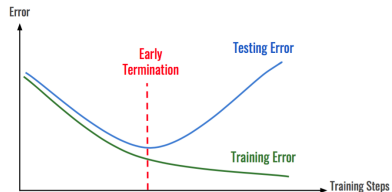


- Only algorithms that learn incrementally have such a plot.

Typical training plots of 'healthy' models

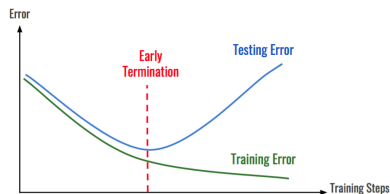


General evolution of a loss function



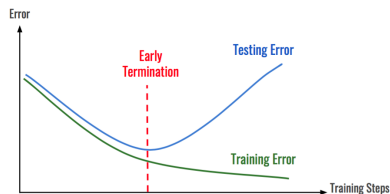
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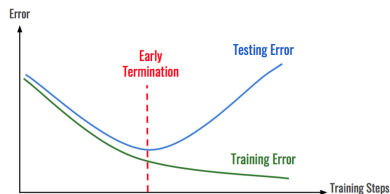
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- Underfitting: Model performs poorly with both datasets, but improves slowly.

General evolution of a loss function



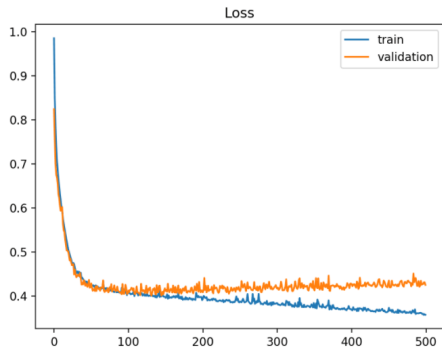
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- Best compromise between memorization and understanding of data.

General evolution of a loss function



- A typical training has three phases:
- Underfitting: Model performs poorly with both datasets, but improves slowly.
- Best compromise between memorization and understanding of data.
- Overfitting: The model memorizes the training data, but can no longer correctly predict the validation data.

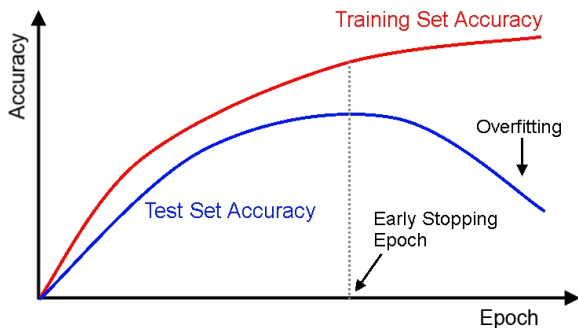
Example of a loss function plot for normal training



- Where does overfitting start?

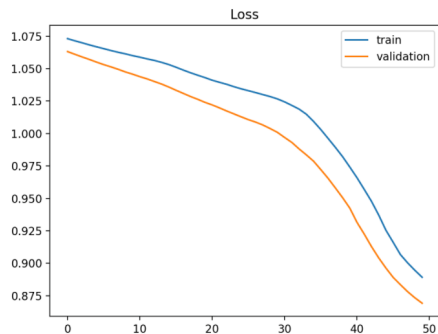
General evolution of a performance metric

- Training should be stopped when the best compromise is reached between memorizing and understanding the data.
- This is the early stopping method.



Different symptoms and their solutions

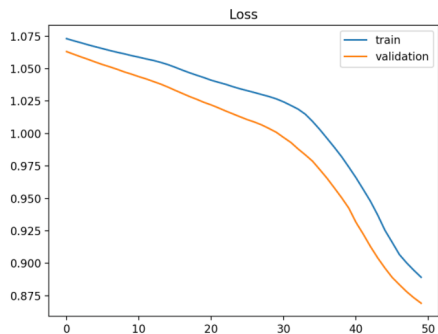
- Training too short



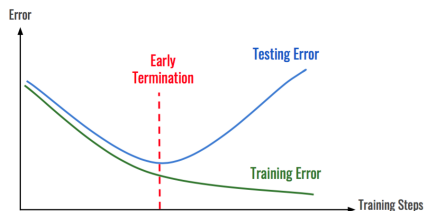
- Training and validation plots have similar behavior and poor performance.

Different symptoms and their solutions

- Training too short

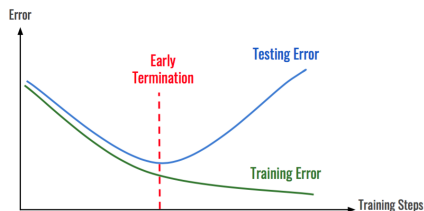
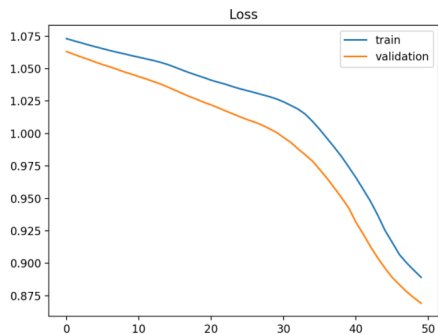


- Training and validation plots have similar behavior and poor performance.



Different symptoms and their solutions

- Training too short



- Training and validation plots have similar behavior and poor performance.
 - Solution: train longer!

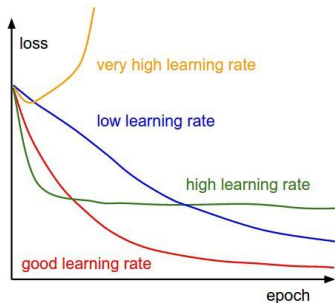
Net well converging learning rate

- In the gradient descent algorithm, the parameters θ of a model are updated according to

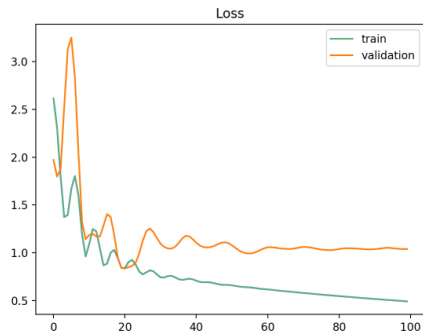
$$\theta = \theta - \eta^* \nabla_{\theta} J(\theta)$$

- where η is the learning rate and $J(\theta)$ is the loss function.
- Learning rate is one of the most important hyperparameters when training a model.

/pause



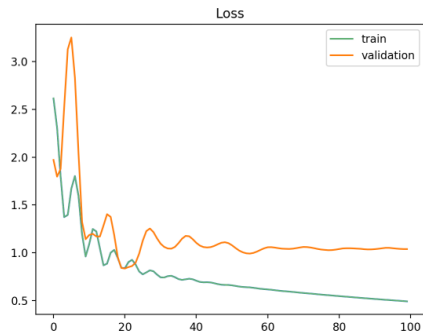
Training set too small



- Iterations

- Training set too small:
- not varied enough in terms of data.
- The trained model recognizes little data from the validation set; He can't make good predictions

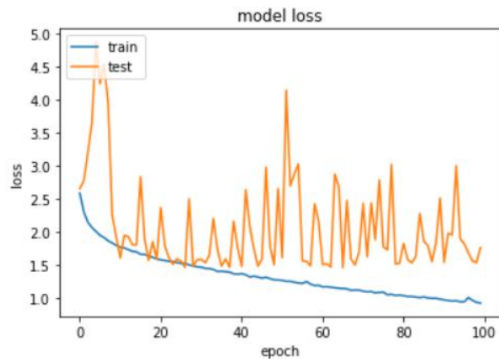
Training set too small



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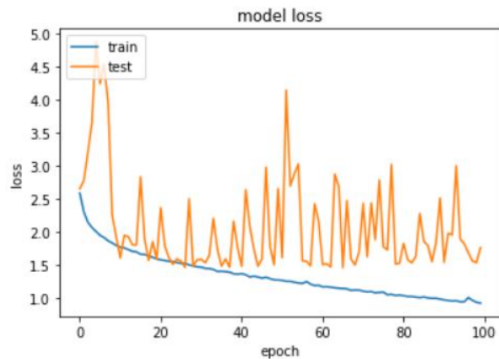
- Training set too small:
- not varied enough in terms of data.
- The trained model recognizes little data from the validation set; He can't make good predictions
- Solutions:
- Increase the number of data
- Standardize the two sets
- Shuffle training data to avoid periodic errors.

Training and validation sets are too different



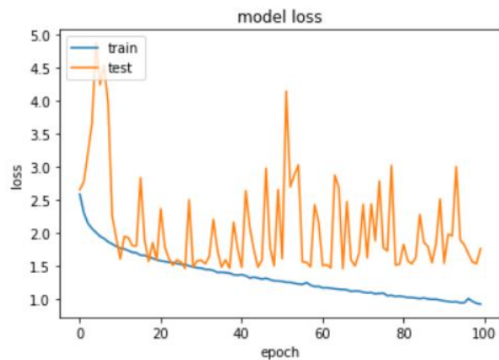
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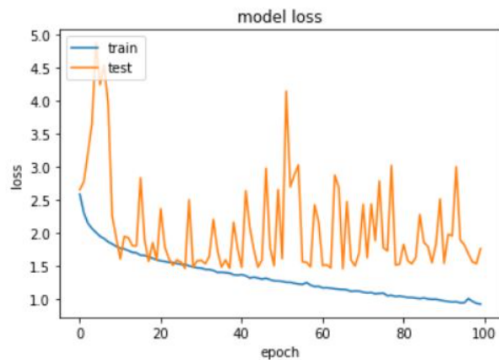
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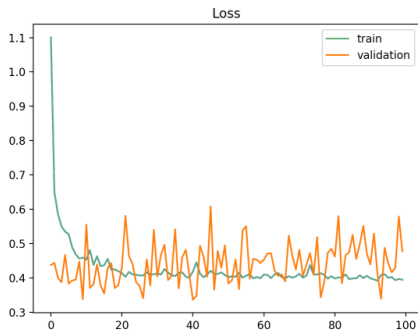
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- Example: The training set contains old data and the validation set contains new data.

Training and validation sets are too different



- The trained model recognizes little data in the validation set;
- It cannot make good predictions.
- Example: The training set contains old data and the validation set contains new data.
- Solutions:
 - Standardize the datasets before separating them into two sets for training and validation.

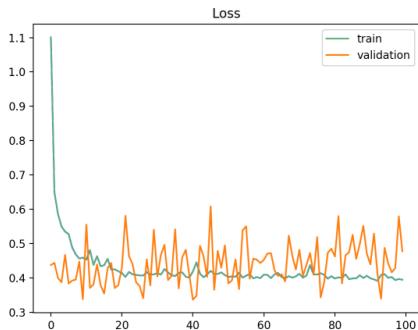
Validation set is too small



- Statistics based on too small a validation set are noisy.
- The two sets may be stratified differently.

- eras

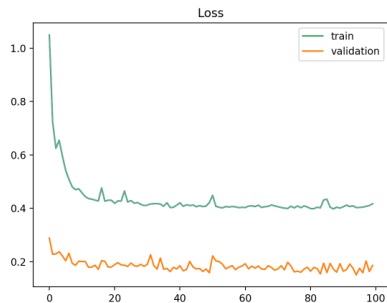
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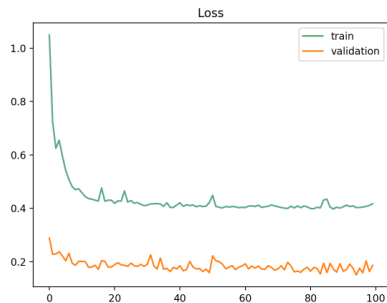
Validation set is too easy



- eras

- Sometimes the validation set contains the most frequent data.
- They are easy to process by the model and few errors are observed.
- Performance in validation is better than in training!

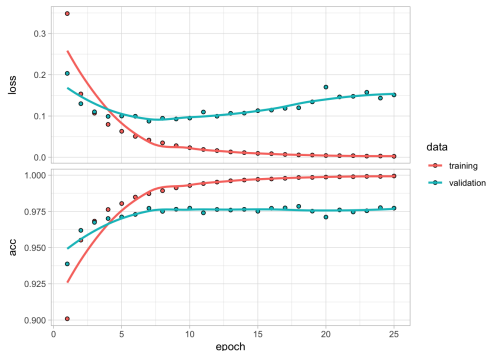
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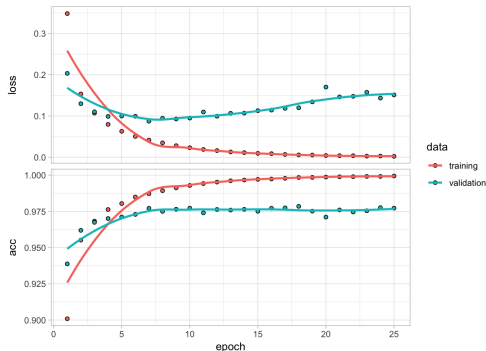
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loss plot or performance metric plot for when to stop?



- Loss function:
 - doesn't always capture what's most important.
 - differentiable function that behaves “similar” to the metric.
 - only for optimization.

loss plot or performance metric plot for when to stop?



- Loss function:
 - doesn't always capture what's most important.
 - differentiable function that behaves “similar” to the metric.
 - only for optimization.
- Performance metric:
 - best defines the performance of the model.
 - depends on the model's intended use.

Performance Metrics

- What is a performance metric?
- A way to quantify the performance of a machine learning model.

Performance Metrics

- What is a performance metric?
- A way to quantify the performance of a machine learning model.
- Note: loss function $\neq$ performance metric
 - The loss function is used during training.
 - The performance metric is used at the end to evaluate the final model.

Different metrics

- Regression
 - R-square
 - Adjusted R-squared
 - MAE
 - MSE
 - RMSE
 - ...
- Classification
 - Exactness
 - Precision
 - Sensitivity
 - Specificity
 - ROC/AUC
 - ...

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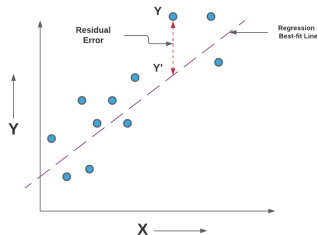
Properties for Metrics

- Metrics are often better evaluations for comparing models
 - We want some function/value that provides a more understandable values
 - Compare across datasets
 - Compare across models
- MSE is not great for this
 - Value are highly sensitive to datasets

Root mean square error (MSE)

- MSE for Mean Squared Error.
- It is the variance of the model's errors.
- It is sensitive to large differences between model predictions and targets.

$$MSE = \frac{1}{N} \sum_{i=1}^N \left(\underbrace{y_i - \hat{y}_i}_{\text{Residual error}} \right)^2$$



Root mean square error (RMSE)

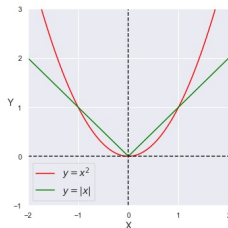
- RMSE for Root Mean Squared Error.
- It is the standard deviation of the errors of the model.
- It is sensitive to large discrepancies between model predictions and observations.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\underbrace{y_i - \hat{y}_i}_{\text{Residual error}} \right)^2}$$

The mean absolute error (MAE)

- MAE for Mean Absolute Error.
- It is robust to large differences between model predictions and observations.
- Reason: the absolute value of the residual error has less effect on the MAE than the **square** of the error on the MSE and RMSE.

$$MAE = \frac{1}{N} \sum_{i=1}^N \left| \underbrace{y_i - \hat{y}_i}_{\text{Residual error}} \right|$$



R-squared coefficient of determination

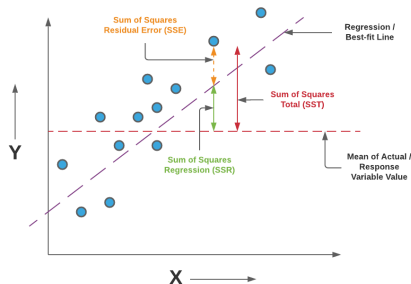
- Calculation method:

If $\bar{y}$ is the mean of the observed data:

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$$

- When $SSE = 0$, data is well explained
- When $SSE = SST$, data is not well explained



- Example for a linear model

R-squared coefficient of determination

- It quantifies the proximity of the data to the fitted model.
- The larger the R^2 value, approaching 1, the more the model explains the variability of the data.

$$R^2 = 1 - \frac{\text{Explained Variation}}{\text{Total Variation}}$$

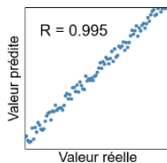
- https://en.wikipedia.org/wiki/Coefficient_of_determination

Examples

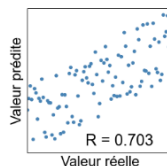
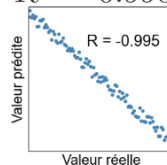
$$R \in [-1, 1]$$

$$R^2 \in [0, 1]$$

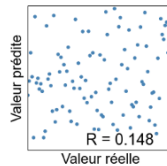
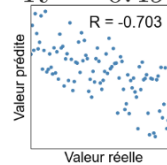
- where R is the correlation coefficient between two sets of data.
- The larger R -squared, the better the match between
- Predictions from a model and observations



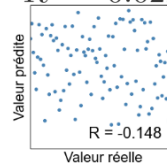
$$R^2 = 0.990$$



$$R^2 = 0.494$$



$$R^2 = 0.022$$



Adjusted R-squared

- The R-squared metric tends to overestimate the quality of the linear regression.
- The adjusted R-squared penalizes for adding additional x-features that do not improve a model.

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$$R^2 = 1 - \frac{SSE}{SST}$$

$$R^2_{\text{ajusté}} = 1 - \left(\frac{N-1}{N-p} \right) \frac{SSE}{SST}$$



$$R^2_{\text{ajusté}} \leq R^2$$

- N: Number of data
- p: Number of parameters of a model

Which metric to choose?

- Typically, the R-squared and a measure of error, e.g. the RMSE.
- The R-squared expresses the percentage of the variance in the data that is explained by a model.
- The RMSE gives the standard deviation of the prediction errors of the model.

Classification

- Metric Selection Depends on Planned Use
- Example of confusion matrix for subjects with Covid-19
- It is assumed that 80% of subjects going to be tested have the disease.
- The figures are reduced to a population of 100 individuals.

100 people at risk of covid-19
Pre-test probability 80%



80 people have covid-19

20 people do not have covid-19

They are tested for covid-19
using the RT-PCR test:



But who actually has covid-19?

57 people have a test result suggesting that
they have covid-19 (test positive)

43 people have a test result suggesting that
they do not have covid-19 (test negative)



Diagnosis

56 people who test
positive have covid-19
("true positive")

1 person who tests positive
does not have covid-19
("false positive")

24 people who test
negative have covid-19
("false negative")

19 people who test negative
do not have covid-19
("true negative")

Consequences

Appropriately told to
self-isolate

Told they need to self-isolate
when they would be safe to
go out

Told they do not need to
self-isolate and so go out and
infect more people

Told they do not need to
self-isolate and are safe to go
out without infecting more
people

Confusion Matrix

- Confusion matrix: explanation of errors made during a binary classification.
- The matrix is not a metric; it helps to calculate several metrics.

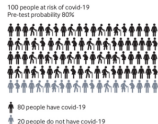


- Number of good classifications: $VP + VN$
- Number of errors: $FN + FP$

	Predicted Positive ($\hat{Y} = 1$)	Predicted Negative ($\hat{Y} = 0$)
Real Positive ($Y = 1$)	True Positives (TP)	False Negatives (FN)
Real Negative ($Y = 0$)	False Positives (FP)	True Negatives (TN)

Metric Selection Depends on Planed Use

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But who actually has covid-19?



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Told they do not need to self-isolate and are safe to go out without infecting more people

		Classe prédite	
		Positif ($\hat{Y} = 1$)	Négatif ($\hat{Y} = 0$)
Vraie classe	Positif ($Y = 1$)	Vrais positifs (VP) 56	Faux négatifs (FN) 24
	Négatif ($Y = 0$)	Faux positifs (FP) 1	Vrais négatifs (VN) 19

- Matching Confusion Matrix

Accuracy

- It is the more intuitive metric.
- It indicates the fraction of correct predictions, among all the predictions made.
- Example: Of the 100 people tested, 75 received a correct diagnosis.

		Classe prédite	
		Positif ($\hat{Y} = 1$)	Négatif ($\hat{Y} = 0$)
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{

$$\begin{aligned}\text{Exactitude} &= \frac{\text{Nombre de bonnes classifications}}{\text{Nombre total d'exemples}} \\ &= \frac{56 + 19}{56 + 19 + 1 + 24} \\ &= 75\%\end{aligned}$$

}

Accuracy

- For some rare diseases, it happens that we measure an accuracy of around 95%.

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- Example for a rare disease:
- 1000 subjects tested: 950 healthy and 50 sick

Accuracy

- For some rare diseases, it happens that we measure an accuracy of around 95%.
- Is this really a good performance in detection?
- **In the case of unbalanced data, accuracy is not a useful metric.**



- Example for a rare disease:
- 1000 subjects tested: 950 healthy and 50 sick
- A bogus test, which always indicates a healthy subject, is right 950 times out of 1000, or 95% accuracy!

Recall

- Recall (or sensitivity): Probability of a positive prediction for a positive case.
- In medicine, it corresponds to the probability of identifying sick patients.
- Example: 80 people with Covid-19 were tested and 56 received a positive diagnosis.

		Classe prédite	
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Vraie classe	Positif ($Y = 1$)	Vrais positifs (VP) 56	Faux négatifs (FN) 24
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$$\begin{aligned}\text{Recall} &= \frac{VP}{VP + FN} \\ &= \frac{56}{56 + 24} \\ &= 70\%\end{aligned}$$

Precision

- Precision: Probability of a positive prediction among the points that were predicted positive.
- In medicine, it corresponds to the probability that patients diagnosed as being sick are really sick.
- Example: 57 people received a positive diagnosis for Covid-19 while only 56 had it.

		Classe prédite	
		Positif ($\hat{Y} = 1$)	Négatif ($\hat{Y} = 0$)
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$$\begin{aligned}\text{Précision} &= \frac{VP}{VP + FP} \\ &= \frac{56}{56 + 1} \\ &= 98\%\end{aligned}$$

Precision or Recall; which one to choose?

- It depends on the needs of the interviewer:

Precision or Recall; which one to choose?

- It depends on the needs of the interviewer:
- We want to minimize false positives $\Rightarrow$ Precision

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Precision or Recall; which one to choose?

- It depends on the needs of the interviewer:
- We want to minimize false positives => Precision
- Example: integrated circuits falsely identified as defective on an assembly line.



- We want to minimize false negatives=> Recall



- Example: patients with undetected cancers.

Specificity

- Specificity: Probability of a negative prediction for a negative case.
- In medicine, it corresponds to the probability of not identifying non-sick patients as sick.
- Example: 20 people without Covid-19 were tested and 19 received a negative diagnosis.

		Classe prédite	
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$$\begin{aligned}\text{Specificity} &= \frac{VN}{VN + FP} \\ &= \frac{19}{19 + 1} \\ &= 95\%\end{aligned}$$

Summary of the main metrics of interest

condition positive (P)

the number of real positive cases in the data

condition negative (N)

the number of real negative cases in the data

true positive (TP)

A test result that correctly indicates the presence of a condition or characteristic

true negative (TN)

A test result that correctly indicates the absence of a condition or characteristic

false positive (FP)

A test result which wrongly indicates that a particular condition or attribute is present

false negative (FN)

A test result which wrongly indicates that a particular condition or attribute is absent

sensitivity, recall, hit rate, or true positive rate (TPR)

$$\text{TPR} = \frac{\text{TP}}{P} = \frac{\text{TP}}{\text{TP} + \text{FN}} = 1 - \text{FNR}$$

specificity, selectivity or true negative rate (TNR)

$$\text{TNR} = \frac{\text{TN}}{N} = \frac{\text{TN}}{\text{TN} + \text{FP}} = 1 - \text{FPR}$$

precision or positive predictive value (PPV)

$$\text{PPV} = \frac{\text{TP}}{\text{TP} + \text{FP}} = 1 - \text{FDR}$$

negative predictive value (NPV)

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} = 1 - \text{FOR}$$

miss rate or false negative rate (FNR)

$$\text{FNR} = \frac{\text{FN}}{P} = \frac{\text{FN}}{\text{FN} + \text{TP}} = 1 - \text{TPR}$$

fall-out or false positive rate (FPR)

$$\text{FPR} = \frac{\text{FP}}{N} = \frac{\text{FP}}{\text{FP} + \text{TN}} = 1 - \text{TNR}$$

false discovery rate (FDR)

$$\text{FDR} = \frac{\text{FP}}{\text{FP} + \text{TP}} = 1 - \text{PPV}$$

false omission rate (FOR)

$$\text{FOR} = \frac{\text{FN}}{\text{FN} + \text{TN}} = 1 - \text{NPV}$$

Classification metrics based on the area under a curve

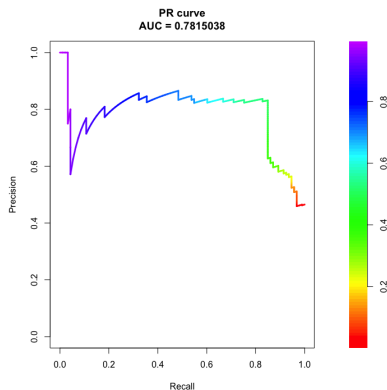
- If we have a model giving the probability of belonging to a class of interest, we can adjust the threshold and classify our data.
- Each threshold leads to a confusion matrix.

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- You can visualize the effect of each threshold using graphs displaying the metrics.
- Area Under Curve (AUC) is a new metric!
- Probability threshold

The ROC curve

- ROC: Receiver Operating Characteristic

The ROC curve

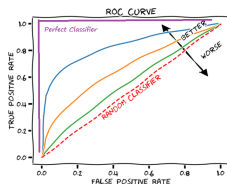
- ROC: Receiver Operating Characteristic
- The AUC represents the probability that a positive vs a negative is decided correctly.

The ROC curve

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- The classifier with the largest area under the curve is the best.

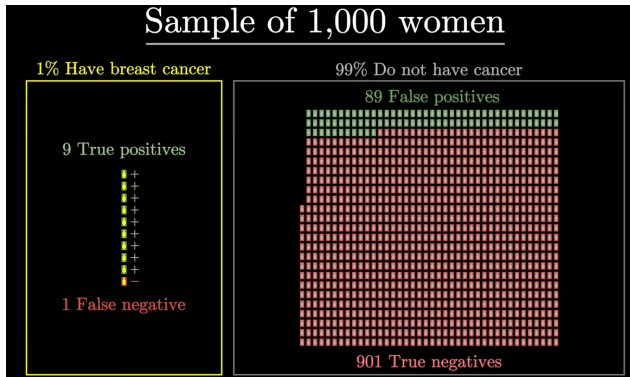
The ROC curve

- ROC: Receiver Operating Characteristic
- The AUC represents the probability that a positive vs a negative is decided correctly.
- The classifier with the largest area under the curve is the best.
- This metric is the most used when unbalancing classes.



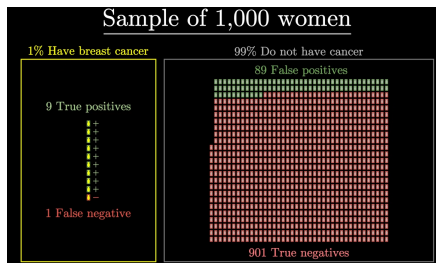
Example of a confusion matrix for subjects with breast cancer

- The test has a sensitivity and specificity of $\sim 90\%$ (source: 3Blue1Brown)



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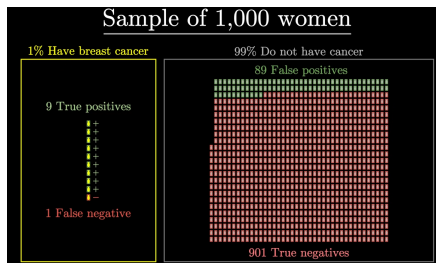
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- Question:
- A 50-year-old woman with no symptoms participates in a routine test and obtains a positive result, what is the probability that she has breast cancer?
- 100%, 90%, 50%, 10%, 0%

Example of a confusion matrix for subjects with breast cancer

- The test has a sensitivity and specificity of ~90%



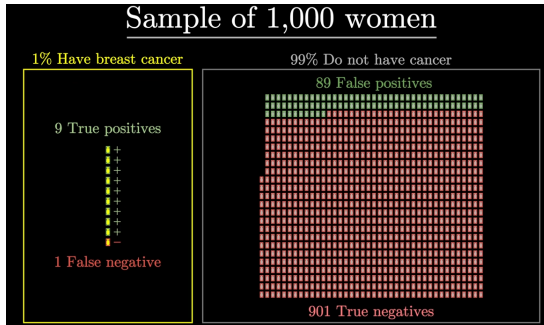
- Question:
- A 50-year-old woman with no symptoms participates in a routine test and obtains a positive result, what is the probability that she has breast cancer?
- 100%, 90%, 50%, 10%, 0%

Answer: - 10%

Example of a confusion matrix for subjects with breast cancer

$$P(\text{Have cancer} \mid \text{positive test}) \approx \frac{9}{9 + 89} \approx \frac{1}{11}$$

- The test has a sensitivity and specificity of ~90%



Conclusion

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- A test determines the probability of being positive

Conclusion

- A test determines whether you are positive or not.
- A test determines the probability of being positive
- A test updates the probability of being positive

Quiz/Poll

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- Use Case: Imagine developing an email service where it is critical to minimize the number of legitimate emails that are incorrectly marked as spam.

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- Use Case: An e-commerce company is developing a system to automatically classify images of its products into broad, well-represented categories, such as “clothing,” “electronics,” “home goods,” and “books.”

Quiz/Poll

- Why is R-squared easier to use than RMSE for evaluating a regression model?
- (A) R-squared is always positive, (B) R-squared is faster to compute, (C) R-squared expresses a percentage, or (D) it's more accurate than giving the standard deviation of the model error

More details on this example and on the statistical tests in the next lessons.

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